TEXORA · SESSIONAL NOTES · STUDY SHEET

Fashion CAD

Working a complete computer-aided design cycle for a garment style — digital pattern drafting, grading, marker planning and a 2D technical flat with specification sheet — and validating the digital output against the manual methods of TE-2207/TE-3108.

AM-4102 · ADVANCED APPAREL MANUFACTURING-I SESSIONAL · STUDY 4 · LEVEL 4 TERM 1

1. Aim and place of the study

Aim: to use a garment CAD system end-to-end — create or digitise a base pattern, add seam allowances and drill marks, grade it across a size range, plan a marker, and produce a technical flat (sketch) with a measurement specification — checking every digital result against hand methods so the software is understood, not merely operated.

CAD in apparel covers four linked work areas: **pattern design (PDS)** — drafting and modifying patterns on screen; **grading** — generating the size set from grade rules; **marker making** — nesting pattern pieces into the most economical lay; and **technical illustration** — flats, spec sheets and 3D visualisation used to communicate the style. The first three replace the manual work of TE-3108 (pattern making, grading, trimming lists) at production speed; the fourth replaces the hand-drawn sketch in the tech pack.

2. Equipment, files and specimens

Equipment: CAD workstation with a garment CAD package (pattern design, grading and marker modules); digitiser or scanner (for entering the paper pattern); plotter/printer for check plots; the physical base pattern and graded nest from TE-3108 Study 6 for cross-checking.

Specimen style: the men's basic trouser of TE-3108 (size M master: waist 82, hip 102, outseam 104, inseam 78 cm; grades: waist/hip 5 cm, outseam 2 cm, inseam 2 cm across S–XL) — the same style whose manual grading is already recorded, so every CAD figure can be audited against a known answer.

3. The CAD workflow

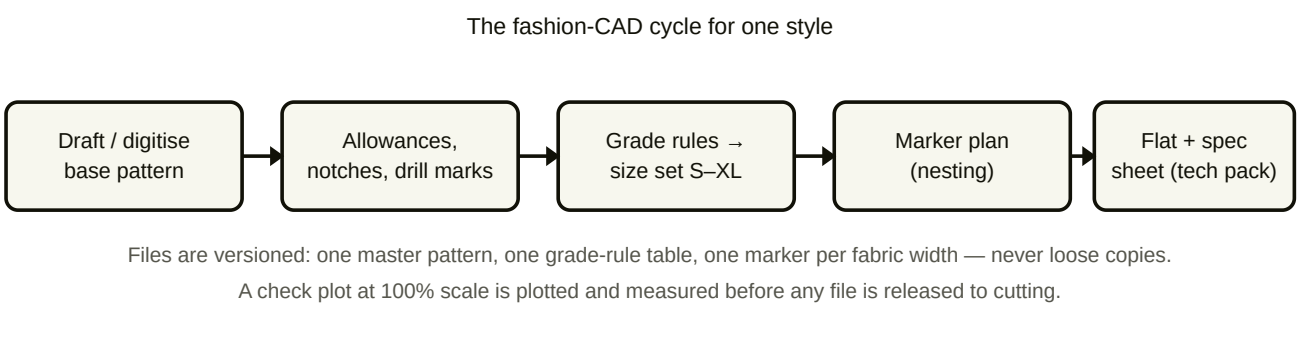


Figure 1 — The CAD cycle. Errors made in step 1 are multiplied by grading and nesting, so verification sits between steps.

MODULE	WHAT IT DOES	TYPICAL ERROR TO GUARD AGAINST
Pattern design (PDS)	Drafting by points, lines and curves; dart manipulation; mirroring; seam allowance generation; internal marks (notches, drills, grain line).	Allowance added to a net line twice; grain line not perpendicular/parallel as specified; curve points too few, giving faceted curves on the plot.
Grading	Each grade point carries X/Y increments per size from the base; the system regenerates every size from the rule table.	Wrong sign/direction on one point (the classic single-point error) — visible only on the graded nest plot.
Marker making	Interactive or automatic nesting of all pieces and sizes within fabric width; reports length and efficiency.	Pieces placed off-grain by rotation shortcuts; efficiency quoted on a marker whose pieces overlap (invalid marker).
Technical illustration / 3D	Vector flats with stitch and trim callouts; spec sheets generated from the pattern's measurement table; 3D drape preview on an avatar.	Flat drawn freehand instead of traced from the actual pattern — proportions drift from the real garment.

4. Worked check — grading and marker figures

VERIFY THE SYSTEM AGAINST TE-3108 HAND RESULTS

Grading: waist grade 5 cm per size, applied as +1.25 cm at each of the four waist grade points (quarter pattern logic of the manual method). CAD nest must read: S 77 · M 82 · L 87 · XL 92 cm at the waist and 97 · 102 · 107 · 112 at the hip — identical to the TE-3108 chart. Any other

reading means a rule error, not a "software difference".
Marker: reference marker for the style (all sizes, 2 garments per size in the lay plan used for costing): length 7.2 m at width 1.45 m → marker area = 7.2 × 1.45 = 10.44 m². Total pattern-piece area reported by the system = 8.62 m² → marker efficiency = 8.62 ÷ 10.44 × 100 = **82.6%**, and the marker carries 6 garments (ratio marker), so fabric consumption = 7.2 ÷ 6 = **1.20 m per garment** before wastage allowances. Both figures must match the cutting-room calculation of AM-4201 Topic 5 — CAD reports, it does not invent.

5. Procedure

1. **Enter the base pattern:** digitise the TE-3108 size-M trouser pattern (front, back, waistband, pocket pieces); verify by plotting at 100% and measuring waist, hip, outseam, inseam against the chart (tolerance ±0.2 cm).
2. **Finish the pattern:** add seam allowances (1 cm general, 1.5 cm side/waistband, 2.5 cm hem — the TE-3108 allowances), notches at hip and knee, drill marks for pocket positions, grain lines and piece names/quantities.
3. **Grade:** build the grade-rule table from the TE-3108 grade increments; generate S, L, XL; plot the graded nest and measure waist/hip of each size against the chart.
4. **Marker:** plan the 6-garment ratio marker at 1.45 m width; record length, efficiency and consumption; compare with §4.
5. **Technical flat and spec:** trace the front/back flat from the pattern outline; add stitch callouts (301 seams, waistband construction, fly) and export the measurement spec table (POM, size M values, tolerances ±, grade increments) as the tech-pack page.
6. **File discipline:** save with the style/version naming used in the lab; archive the check plots with initials and date.

Observation points — what to record

- Digitising accuracy: measured plot values vs chart, per POM.
- The grade-rule table as entered, and the nest check for every size — including any single-point error found and fixed.
- Marker length, efficiency and consumption vs the §4 reference figures, with reasons for any difference (width, ratio, piece count).
- What the flat/spec export drew from the pattern automatically vs what had to be added by hand.

Precautions

- Always plot and physically measure a check plot before release: screen scale lies, paper does not.
- One master file per style — parallel copies ("final2") are how factories cut the wrong revision.
- Back up the grade-rule table with the pattern: a pattern without its rules cannot be re-graded after edits.
- Check units (cm vs inch) at file import — the classic silent CAD disaster.

Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why is the check plot measured physically when the CAD system already reports measurements?
2. A graded nest shows L waist 86 cm instead of 87. Where is the error, and why does it appear in only one size?
3. State the marker efficiency formula and compute it for a marker of length 6.5 m, width 1.40 m, total piece area 7.55 m².
4. Why must the technical flat be traced from the pattern rather than drawn freehand?
5. Give two advantages of CAD grading over manual grading beyond speed.

Answer key

1. Digitising and rule entry are human steps with silent errors (scale, units, a mis-typed increment). The plot converts the file back into a physical object that can be measured with a tape — the only independent check that the file describes the intended garment.
2. In the grade-rule table: one waist grade point carries a wrong increment (e.g. 1.0 instead of 1.25 cm, or a wrong sign at one point). Only size L is wrong because each size is generated from the base by its own cumulative increments — an error in the L step does not touch S or XL unless their rules share the wrong entry.
3. Efficiency = (total pattern-piece area ÷ marker area) × 100. Marker area = 6.5 × 1.40 = 9.10 m². Efficiency = 7.55 ÷ 9.10 × 100 = 83.0%.
4. The flat is a contract document: buyers and factories read proportions, pocket positions and construction from it. Traced from the pattern, it is guaranteed to describe the garment that will actually be cut; freehand drawing quietly substitutes the illustrator's idea for the pattern.
5. Consistency (every size is generated by the same rules, so no size drifts) and editability with control (change the base once and regenerate the whole set, with the nest plot exposing single-point errors immediately). A third: the same file feeds marker making and spec tables, removing transcription errors between departments.